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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	UM IS ONE
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 – Quality Education
Project / Solution Name	AI Skills Compass
Name (Leader)	ANG JI JIN
Email & Phone (Leader)	angjijin20071008@gmail.com & 017-823 9621
University Affiliation(s)	University Malaya
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is reshaping what Malaysian jobs require faster than workers, students and employers can adapt. The WEF Future of Jobs Report 2025 expects 39% of workers' core skills to change by 2030; of every 100 workers, 59 will need training, and 11 of them are unlikely to receive it. Malaysia already has a skills mismatch: DOSM Labour Force Survey data show about **1.95 million employed Malaysians in skills-related underemployment (≈11.5% of 16.87 million employed), including about 476,000 aged 15–24.**

Who is affected: mid-career workers in AI-exposed roles such as bank tellers, clerks and data-entry staff; fresh graduates and TVET students choosing courses without current market signals; and SMEs and HR teams that cannot afford workforce-planning consultants.

Why it matters: economically, underemployment wastes education spending and holds down wages and productivity. Socially, the people with the least access to career guidance (women returning to work, B40 households, rural learners and non-English speakers) are the most exposed, so without help inequality widens. Today career advice relies on static reports and opinion rather than live data, so workers cannot see which skills are rising or falling, or which free courses would close their gap.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AI **Skills Compass** is a web application that takes a person from who they are today to where they can be in the AI era, in five steps:

- **Skills profile:** the user uploads a CV, fills in a guided form or picks a sample profile. An AI model extracts skills and shows the CV phrase behind each one; the user confirms the result.
- **AI-trend signal layer:** a −100 to +100 trend index for 39 skills, combining the WEF 2025 baseline, Hacker News discussion momentum and skill demand in live job advertisements, including 2,325 Malaysian JobStreet postings our team collected across 10 sectors. Snapshots are dated and refreshed weekly.
- **Explainable role matching:** 20 future roles ranked by skills fit, outlook and interest, under slow, moderate and fast AI-disruption scenarios; roles in the top 5 in all three are flagged "robust".
- **Reskilling pathway:** a week-by-week plan built from 38 real free or low-cost courses, with HRD Corp and MyFutureJobs next steps.
- **SDG mapping and HR view:** every skill and role is linked to SDGs with a stated reason; employers get a team skills heatmap.

How AI is used: GenAI and ML models read CVs, forecast skill trends and explain results in plain English, Bahasa Malaysia or Chinese. Scores always come from a transparent, unit-tested formula, never from a model, so recommendations are auditable. Protected attributes are never collected, and an automated fairness test confirms rankings do not change when they are added.

Workshop skills applied: GenAI APIs (an LLM served by vLLM on RunPod Serverless plus custom skill-extraction and forecasting workers); database architecture (a versioned, dated snapshot store, JSON taxonomies and a Python ETL pipeline for JobStreet data); and cloud hosting (Replit Autoscale deployment and a weekly GitHub Actions data refresh).

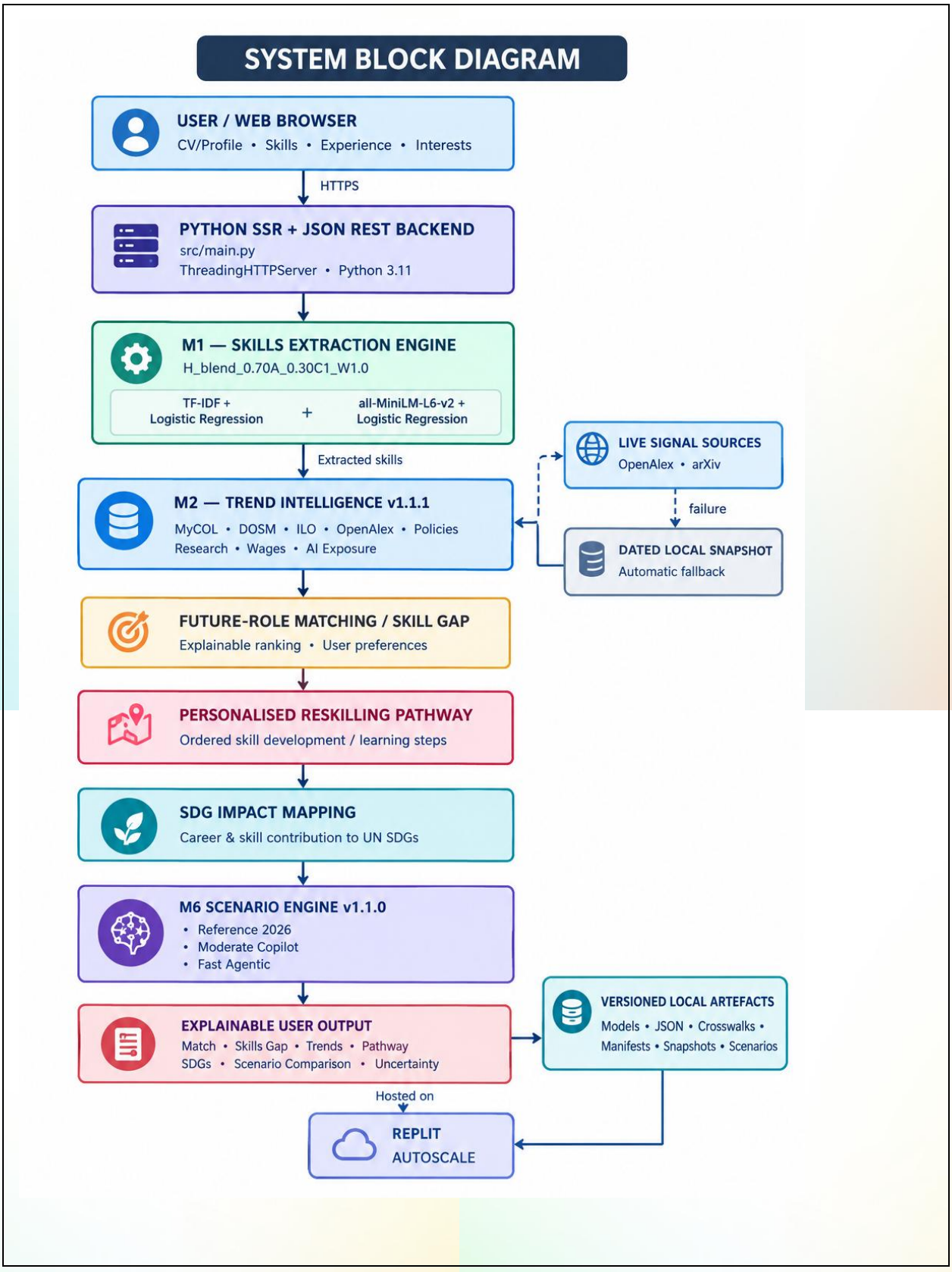
3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology
Frontend	HTML/CSS/vanilla JavaScript single-page app (~105 KB, no framework, mobile and low-bandwidth friendly); pdf.js for PDF CVs; English / Bahasa Malaysia / Chinese interface
Backend	Node.js 20 zero-dependency server: static hosting, /api/refresh, /api/status, /api/ai/* endpoints, per-IP rate limiting
Scoring engine	engine.js: pure, unit-tested functions for the trend index, match score, pathway and SDG weighting
AI / GenAI	RunPod Serverless: (1) CV skill-extraction worker, (2) trend-forecasting worker, (3) vLLM-hosted LLM (e.g. Llama-3.1-8B-Instruct) for plain-language explanations; each has a labelled fallback
Data sources	WEF Future of Jobs 2025; Hacker News Algolia API; Arbeitnow and Remotive job APIs; data.gov.my (DOSM Labour Force Survey); JobStreet Malaysia postings (Python collector + build_dataset.py ETL)
Data store	Versioned, dated JSON trend snapshots plus curated taxonomies (39 skills, 20 roles, 38 courses, SDG links)
DevOps	GitHub; GitHub Actions (weekly data refresh); Replit Autoscale deployment; node --test (19 tests incl. a fairness test)


SYSTEM BLOCK DIAGRAM
USER / WEB BROWSER
CV/Profile • Skills • Experience • Interests
HTTPS
PYTHON SSR + JSON REST BACKEND
src/main.py
ThreadingHTTPServer • Python 3.11
M1 — SKILLS EXTRACTION ENGINE
H_blend_0.70A_0.30C1_W1.0
TF-IDF + Logistic Regression
+
all-MiniLM-L6-v2 + Logistic Regression
Extracted skills
LIVE SIGNAL SOURCES
OpenAlex • arXiv
failure
DATED LOCAL SNAPSHOT
Automatic fallback
M2 — TREND INTELLIGENCE v1.1.1
MyCOL • DOSM • ILO • OpenAlex • Policies
Research • Wages • AI Exposure
FUTURE-ROLE MATCHING / SKILL GAP
Explainable ranking • User preferences
PERSONALISED RESKILLING PATHWAY
Ordered skill development / learning steps
SDG IMPACT MAPPING
Career & skill contribution to UN SDGs
M6 SCENARIO ENGINE v1.1.0
• Reference 2026
• Moderate Copilot
• Fast Agentic
EXPLAINABLE USER OUTPUT
Match • Skills Gap • Trends • Pathway
SDGs • Scenario Comparison • Uncertainty
VERSIONED LOCAL ARTEFACTS
Models • JSON • Crosswalks •
Manifests • Snapshots • Scenarios
Hosted on
REPLIT
AUTOSCALE

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we built personas for the groups hit hardest: a bank teller in an AI-exposed role, a career switcher, a student, and an HR manager with an eight-person team. Their pain points: “I don’t know which skills will matter”, “courses are expensive”, “advice isn’t in my language” and “I don’t trust a black-box score”.

Define: a Malaysian worker needs a trustworthy, free, personalised path to a growing role without needing a career counsellor.

Ideate: we compared a chatbot advisor, a job board and a static report, then chose explainable scoring with GenAI used only for language tasks, so users see why and models cannot invent jobs or courses.

Prototype and test: a deployed app with three onboarding paths, editable CV results, scenario ranges instead of one prediction, three languages and a low-bandwidth build; 19 automated tests check the engine, data sources and fairness.

Phase 2 readiness: the app runs on Replit Autoscale, data refreshes weekly, every model has a fallback, rate limiting protects cloud credit and profiles stay in the browser. Next: Malaysian job data in the live signal and a pilot with university and TVET career centres.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary – Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." **Secondary – Target 8.5:** "Achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities…"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**≈11.5% of employed Malaysians (≈1.95 million people, ≈476,000 aged 15–24) are in skills-related underemployment** [2]. Globally, 39% of core skills are expected to change by 2030, and 11 in every 100 workers who need reskilling are unlikely to receive it [1]. In our prototype, 0% of the live job-demand signal is Malaysian today; 2,325 local JobStreet postings have been collected to change this. Sources: [1] World Economic Forum (2025), The Future of Jobs Report 2025, weforum.org/publications/the-future-of-jobs-report-2025. [2] Department of Statistics Malaysia, Labour Force Survey (lfs_month, Aug 2026; lfs_qtr_sru_age, Q2 2026) via data.gov.my/data-catalogue, retrieved 19 Oct 2026; rate = 1,948.2k ÷ 16,874.4k employed. [3] ILO (2023), Generative AI and Jobs: A global analysis (AI-exposure assumptions).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**1. Skills-gap closure:** % increase in a user's skills-fit score toward their chosen future role, measured in-app before the pathway and after the first plan milestones. **2. Plan-to-action rate:** % of users who start at least one recommended free course within 14 days.

Metric Category	Your Response
	3. Technical KPI: precision (%) of AI skill extraction against a hand-labelled sample of Malaysian CVs and JobStreet postings.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Pilot with ~50 Malaysian students, workers and HR users: • **≥20% average relative improvement in skills-fit score** toward the chosen role (e.g. 55 → 66). • **≥60%** of pilot users start a recommended course within 14 days. • **≥80%** precision in CV skill extraction. • **≥50%** of job-demand signals from Malaysian postings (from 0% today).

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Category	Third-Party Resource	Use / Attribution
Open-source library	*scikit-learn*	ML preprocessing, classification and evaluation; BSD-licensed
Open-source library	*NumPy*	Numerical computing
Open-source library	*pandas*	Dataset processing
Open-source library	*sentence-transformers*	Sentence-embedding framework
Open-source dependency	*PyTorch / Hugging Face Transformers*	MiniLM inference dependencies
Pre-trained model	*sentence-transformers/all-MiniLM-L6-v2*	Semantic sentence embeddings; model attribution retained
Public taxonomy	*MASCO 2020*	Malaysian occupational classification; Ministry of Human Resources
Public labour-market dataset	*TalentCorp MyMahir MyCOL*	Malaysian critical-occupation evidence; TalentCorp
Public labour-market data	*OpenDOSM / DOSM*	Labour-force and wage evidence
Public occupational taxonomy	*ESCO*	European Commission occupations and skills classification

Category	Third-Party Resource	Use / Attribution
Public research data	*OpenAlex*	Scholarly research metadata and trend signals
Public research service	*arXiv*	Research publication metadata
Public research publication	*ILO WP96 / WP140*	GenAI occupational exposure evidence
Public policy documents	*MOSTI, Bursa Malaysia, JPDP, MyDIGITAL and MITI publications*	Malaysian AI, sustainability, privacy and industry policy evidence

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



October 24 2026